Lecture 21 (March 21)

Again, we will have a 'free extension' on the problem set through the weekend.

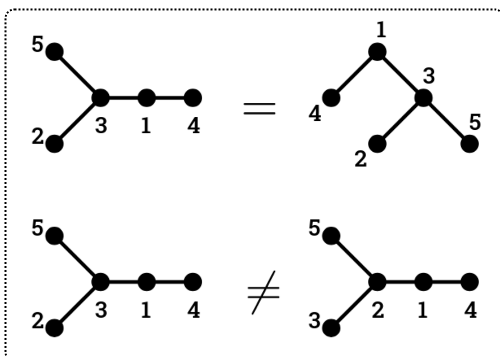
Definition (Labeled trees).

A **labeled tree** $T = (V, E)$ is a connected simple graph without cycles on the vertex set $\{1, 2, 3, \ldots, n\}$.

(Simple graph means that at most one edge connects any two vertices, and no edges connect a vertex to itself.)

The structure of the tree is just the set of edges; the way it is drawn on the plane does not matter.

For example:



A **leaf** in T is a vertex of degree 1. We have a couple of simple lemmas.

Lemma 1 (Leaves).

Any tree T on $n \geq 2$ vertices has ≥ 2 leaves.

Lemma 2 (Edges).

Any tree on n vertices has exactly $n - 1$ edges.

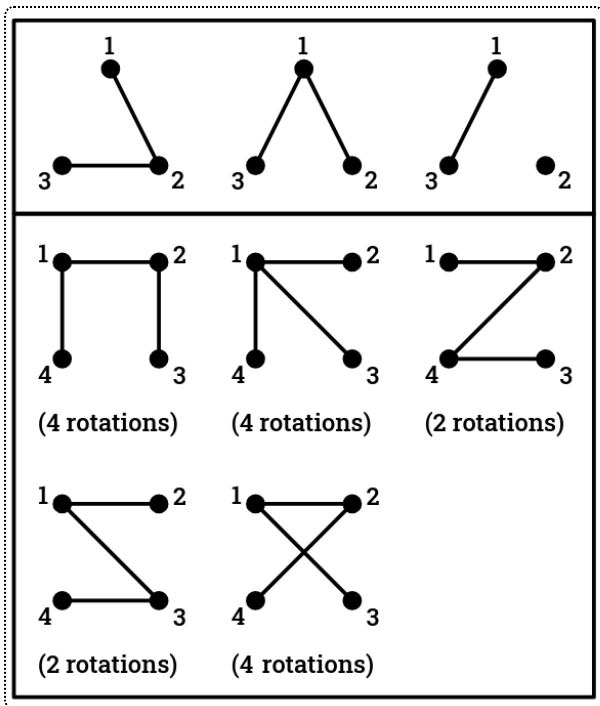
Cayley's Formula

Theorem 3 (Cayley's Formula).

There are n^{n-2} labeled trees on n verties.

A generalization of this theorem was proven in a 1889 paper of Cayley, but it was proven earlier by Borchardt in 1860 and Sylvester in 1857.

As an example, here are all 3^1 labeled trees with 3 vertices and 4^2 labeled trees with 4 vertices.

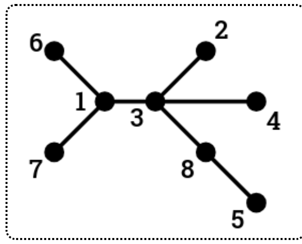


Our goal today is to prove Cayley's Theorem. It's hard to use induction since the expression n^{n-2} doesn't clearly relate to the expression for $n-1$.

So, we first generalize it. Let $x_1, x_2, \ldots, x_n$ be variables. For a labeled tree T on n vertices, denote

$$x^T := \prod_{i=1}^{n} x_i^{\deg_T(i)-1}.$$

For example, the following tree T has $x^T = x_1^2 x_3^3 x_8$.



Then:

Theorem 4 (Multivariate Cayley's Formula).

$$\sum_{|T|=n} x^T = (x_1 + x_2 + \cdots + x_n)^{n-2},$$

where the sum is over all labeled trees on n vertices.

Clearly, plugging in $x_1 = x_2 = \cdots = 1$ yields Cayley's Formula.

Proof. Let $g_n(x_1, \ldots, x_n) := \sum_T x^T - (x_1 + x_2 + \cdots + x_n)^{n-2}$. We want to show that $g_n = 0$; induct on n. The base case $n = 2$ is clear: $1 - 1 = 0$.

Here are some observations:

1. g_n is a polynomial in n variables.
2. $\sum_T x^T$ has degree $2(n-1) - n = n - 2$ (by counting edges), so g_n has degree at most $n - 2$.
3. If we plug in e.g. $x_n = 0$, then $\sum_T x^T$ will only count trees where n is a leaf. For any labeled tree T' on $n - 1$ vertices, there is one such tree for each vertex that n could connect to, so:

$$\begin{aligned} g_n(x_1, \ldots, x_{n-1}, 0) &= \left(\sum_{\substack{T' \text{ on} \\ n-1 \text{ vtxs}}} x^{T'} \right) (x_1 + x_2 + \cdots + x_{n-1}) - (x_1 + \cdots + x_{n-1} + 0)^{n-2} \\ &= (x_1 + \cdots + x_{n-1}) \left(\left(\sum_{|T'|=n-1} x^{T'} \right) - (x_1 + \cdots + x_{n-1})^{n-3} \right) \\ &= (x_1 + \cdots + x_{n-1}) \cdot g_{n-1}(x_1, \ldots, x_{n-1}) \\ &= 0. \end{aligned}$$

Note that observation 3 applies when setting any $x_i = 0$, not just x_n. We claim that these three facts are enough to prove the theorem.

Lemma 5 (Polynomial is identically 0).

For any polynomial $g(x_1, \ldots, x_n)$ such that $\deg(g) < n$ and $g|_{x_{i=0}} = 0$ for all $i = 1, \ldots, n$, we have $g(x_1, \ldots, x_n) = 0$.

Proof. Suppose not. Write g as the sum of monomials $\sum \alpha_{(a_1,\ldots,a_n)} x_1^{a_1} \cdots x_n^{a_n}$. Then, one of the monomials is nonzero, so for some $(a_1, \ldots, a_n)$, $\alpha_{(a_1,\ldots,a_n)} \neq 0$. But since the degree is less than n, we have $a_i = 0$ for some i. Then $g|_{x_i=0} \neq 0$ (since it will have this monomial as a nonzero term), a contradiction.

By this lemma, we are done!

This proof seems slightly suspicious since we barely used any properties of labeled trees, but it is correct.

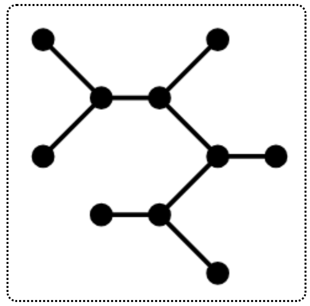
Corollary 6 (Restricting degrees).

The number of labeled trees T on n verties such that $\deg_T(i) = d_i$ for all $i = 1, \ldots, n$ is

$$\begin{aligned}[x_1^{d_{n-1}} x_2^{d_{n-2}-1} \ldots x_n^{d_{n-1}}](x_1 + \cdots + x_n)^{n-2} &= \binom{n-2}{d_1 - 1, d_2 - 1, \cdots, d_n - 1} \\ &= \frac{(n-2)!}{(d_1-1)!(d_2-1)!\cdots(d_n-1)!}\end{aligned}$$

(Recall the $[\cdot]$ notation means "the coefficient of".)

For example, let's count 3-valent (trivalent) trees, where all non-leaf vertices have degree 3, like the following:



Note that there are always two more leaves than non-leaves (and an even number of total vertices). So WLOG suppose $1, \ldots, m+1$ are leaves and $m+2, \ldots, 2m$ have degree 3.

The number of such trees is $\frac{(2m-2)!}{2^{m-1}}$ by the above corollary, so the total number of trivalent trees on $2m$ vertices is

$$\frac{(2m-2)!}{2^{m-1}} \cdot \binom{2m}{m-1}.$$

Lecture 23 (March 31)

(Unfortunately, I was not in class today; thanks to Annie Guo for sending notes! These notes are also based on notes from 2021 published by Prof. Postnikov.)

Today we will discuss two bijective proofs of Cayley's formula (in addition to the algebraic one from last class). Recall that the formula says that there are n^{n-2} labeled trees on n vertices.

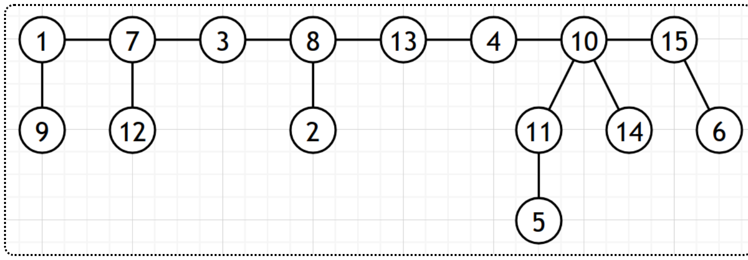
Proof 1 (Eğecioğlu and Remmel, 1986)

The first proof is due to Eğecioğlu and Remmel, based on Joyal 1981.

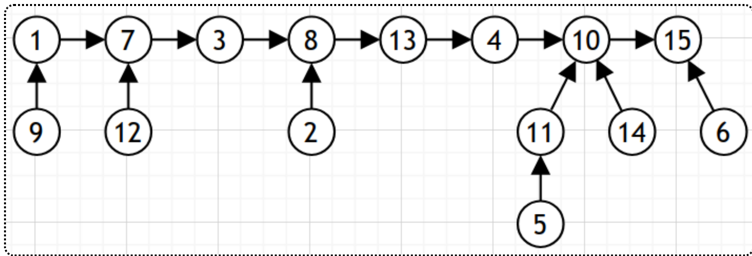
We will express a bijection from labeled trees T on n vertices to maps $f\colon [n] \to [n]$ such that $f(1) = 1$ and $f(n) = n$. There are clearly n^{n-2} of these. We demonstrate this with an example.

Forward map

Suppose $n = 15$. We draw our tree so that the path P from 1 to n is arranged in a line, and other vertices are connected below, like so:

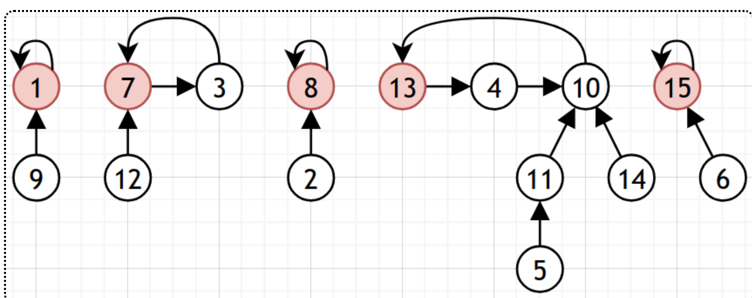


Then, we direct all edges towards n:



Our goal is to make this directed graph represent f, so that k points to $f(k)$. Every edge except n has outdegree exactly 1.

We mark the records in the path from 1 to n (like we would with a permutation). In a similar transformation to the one we used to prove cyc and rec were equidistributed, we disconnect all edges before records and change them into cycles instead:



This is our map.

$$\begin{array}{ccccc} 1 \to 1 & 2 \to 8 & 3 \to 7 & 4 \to 10 & 5 \to 11 \\ 6 \to 15 & 7 \to 3 & 8 \to 8 & 9 \to 1 & 10 \to 13 \\ 11 \to 10 & 12 \to 7 & 13 \to 4 & 14 \to 10 & 15 \to 15 \end{array}$$

Note that indeed, we always end up with loops $1 \to 1$ and $15 \to 15$.

For the inverse map, we draw the directed graph for f, align all the cycles in increasing order of maximal element, rotate them so that the maximal elements are first, and rejoin the path from 1 to n. This always results in the original labeled tree (and is possible since $f(1) = 1$ and $f(n) = n$.)

In fact, it is possible to prove stronger results by looking at other statistics on our trees.

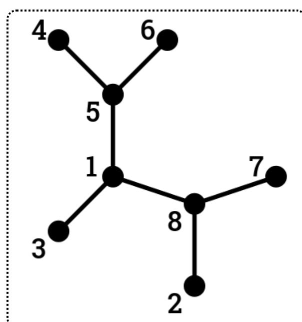
Proof 2 (Prüfer codes)

We will map every labeled tree T to a sequence $\mathrm{code}(T) = (c_1, \ldots, c_{n-2})$ of $n-2$ numbers in $[n]$. These sequences are known as **Prüfer codes**.

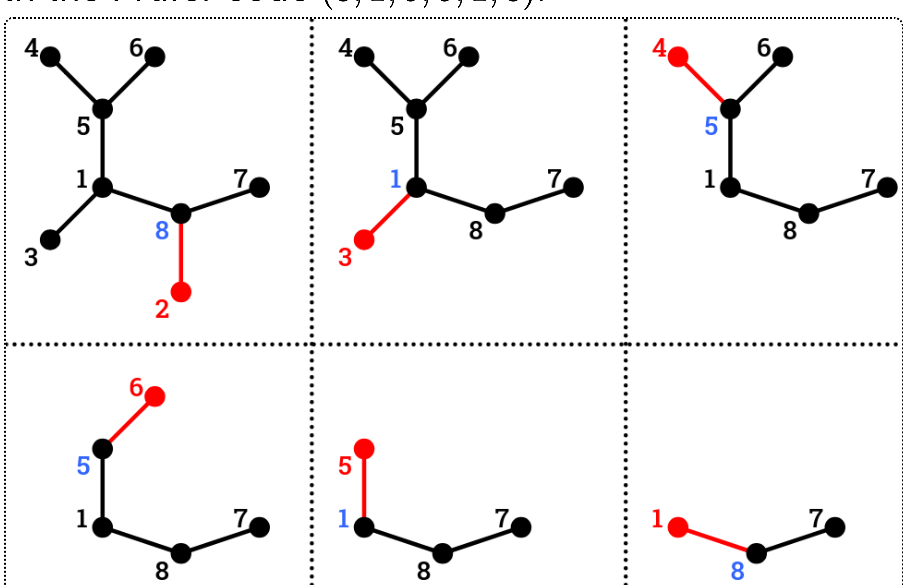
This is done through a process of repeatedly removing a leaf from the tree, namely:

- Find the leaf with minimal label ℓ. Suppose its unique edge is (ℓ, c).
- Record c and erase the edge (ℓ, c).
- Repeat these steps $n-2$ times, until there is only one edge left.

For example, given the following tree,



we end up with the Prüfer code $(8, 1, 5, 5, 1, 8)$:



For the inverse construction, we use the following fact:

Theorem 7 (Degree of vertices in Prüfer codes).

For all i in $[n]$,

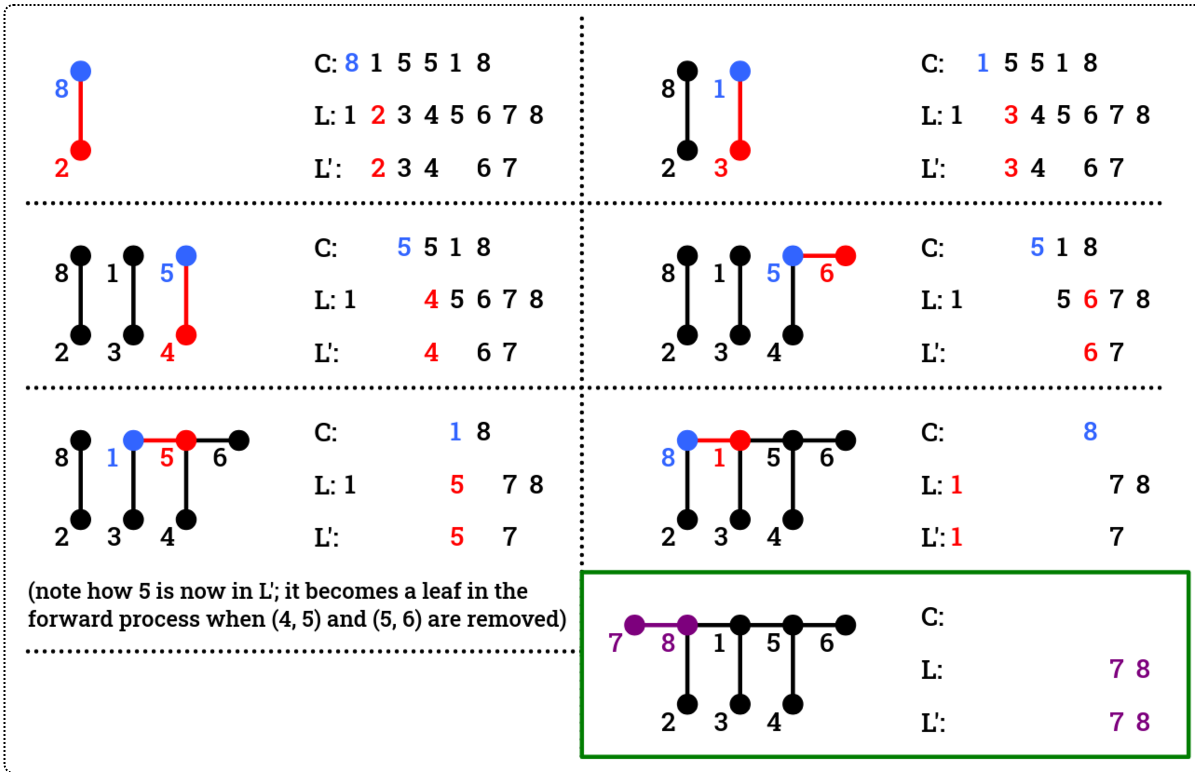
$$\#\{j \mid c_j = i\} = \deg_T(i) - 1.$$

For example, we can immediately retrieve that the leaves in our original tree are $2, 3, 4, 6, 7$, since they don't appear in the Prüfer code. The proof is that i is recorded whenever any edge incident to it is removed, except for the last one (when i is a leaf).

As such, we can reconstruct the tree in order, keeping track of the remaining portion C of the Prüfer code and the set of vertices L that still have unadded edges. At the beginning, $C = \mathrm{code}(T)$ and $L = [n]$.

- At every step, we want to maintain the property that the vertices that still need exactly one more edge (i.e. those that would be leaves at the same step in the removal process) are exactly those that appear in L but not C. Let $L' = L \setminus C$ (though C is a sequence, not a set).
- When we parse the first number c in C, we add an edge (c, ℓ), where $\ell = \min(L')$. This is the same edge that we removed at this step in the forward process: L' consists of all current leaves after previous edge deletions, so ℓ is the minimum leaf, and we recorded c as its neighbor.
- ℓ only needed one more edge, which we just added, so we can remove it from L. We move onto the next step by removing c from C; this may or may not add c to L' (depending on whether it's recorded again).
- Repeat these steps $n - 2$ times. C will be empty, and there will be two remaining elements in L, which we connect for the last edge.

Here's our example:



This adds edges in the same order that we removed them, so we end up with the original T and this is a bijection as desired.

Lecture 24 (April 2)

The theorem used in Proof 2 above has a corollary:

Corollary 8 (Degree restrictions).

Given fixed $d_1, \dots, d_n \geq 1$ such that $\sum d_i = 2(n-1)$, the number of labeled trees T on n vertices such that $\deg_T(i) = d_i$ for all i is

$$\binom{n-2}{d_1 - 1, d_2 - 1, \dots, d_n - 1} := \frac{(n-2)!}{\prod_i (d_i - 1)!}.$$

As an example, if $(d_1, \dots, d_n) = (1, 2, \dots, 2, 1)$, the tree must be some path from 1 to n, so the number of trees is just the number of permutations of $2, 3, \dots, n-1$, i.e. $(n-2)!$.

Spanning Trees

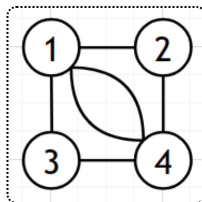
Suppose we have a graph $G = (V, E)$, allowing multiple edges, loops, etc.

Definition (Spanning trees).

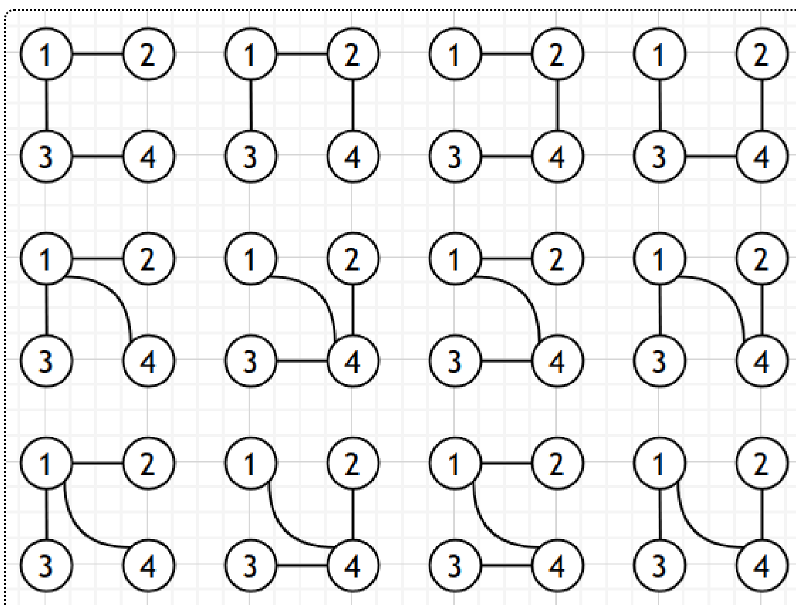
A **spanning tree** in G is a subgraph $T = (V, E')$ where $E' \subseteq E$, such that T is a tree.

Let $t(G)$ denote the number of all spanning trees of G. For example, if G is disconnected, $t(G) = 0$.

Here's our example graph G (note that there are two edges from 1 to 4).



$t(G) = 12$, as shown:



Now, we can reformulate Cayley's Formula as saying that

$$t(K_n) = n^{n-2}.$$

What do we think $t(K_{m,n})$ (a complete bipartite graph, where all m vertices on the left are connected to all n vertices on the right) is? It should be symmetric in m and n and look vaguely similar to n^{n-2}.

In fact, $t(K_{m,n}) = m^{n-1} \cdot n^{m-1}$.

What about $t(K_{m,n,k})$? That will be left as an exercise on the next problem set.

We can come up with a multivariate version of the formula for $t(K_{m,n})$, like how we had a multivariate version of Cayley's Formula.

Given variables $x_1, \cdots, x_m, y_1, \cdots, y_m$ and a tree $T \subset K_{m,n}$, we define

$$\mathrm{wt}(T) := \prod_{i=1}^{m} x_i^{\deg_T(i)-1} \cdot \prod_{j=1}^{n} y_i^{\deg_T(j')-1}$$

(This is essentially the same polynomial from multivariate Cayley's Formula.)

Theorem 9 (Multivariate $t(K_{m,n})$).

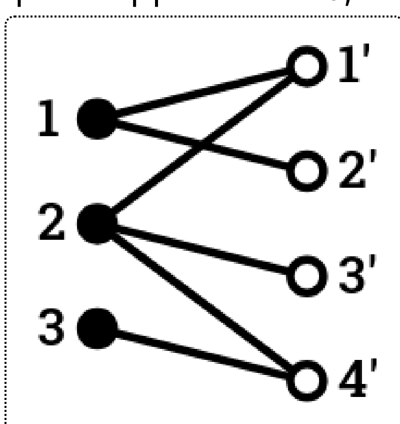
$$\sum_{T \subset K_{m,n}} \mathrm{wt}(T) = (x_1 + \cdots + x_m)^{n-1} \cdot (y_1 + \cdots + y_n)^{m-1},$$

where the sum is over all spanning trees of $K_{m,n}$.

Let's try to find $t(K_{m,n})$ bijectively. We can invent a bipartite version of Prüfer coding.

Given $T \subset K_{m,n}$, we define its bipartite Prüfer code $(a_1, \ldots, a_{n-1}; b_1, \ldots, b_{m-1})$ such that $a_i \subset \{1, \ldots, m\}$ and $b_j \subset \{1', \ldots, n'\}$.

Once again, let's do an example. Suppose $m = 3$, $n = 4$, and this is our tree:



Assume that $1 < 2 < 3 < 1' < 2' < 3' < 4'$ (since Prüfer coding requires us to define an ordering).

In the usual Prüfer code, we would repeatedly remove the minimal leaf and record its adjacent vertex, until we have only one edge left. This would yield $(4', 1, 1', 2, 2)$.

In order to modify this procedure to obtain its bipartite code $\mathrm{code}^{\mathrm{bip}}(T)$, we simply move all of the primed numbers (those with ') to the right. So we'd get $(1, 2, 2; 4', 1')$. The claim is that this is a bijection, so we need to construct the reverse map. This will once again be left as an exercise on the problem set.

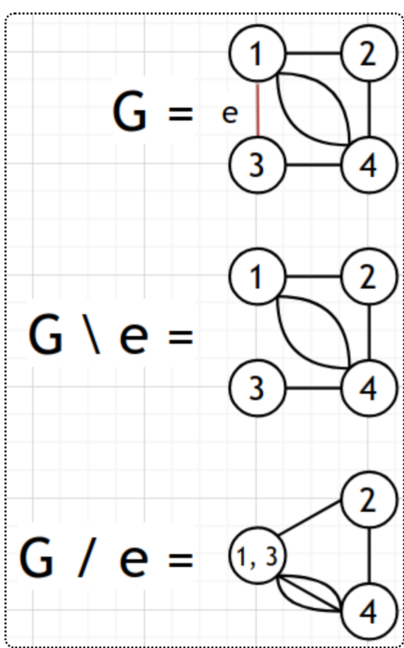
Deletion-contraction

One way to count the number of spanning trees in an arbitrary graph is through "deletion-contraction".

Consider a graph $G = (V, E)$, not necessarily simple. Suppose $e \in E$ is not a loop (it connects two distinct vertices.)

There are two operations we can do on this graph: deletion $G \setminus e$ (removing the edge) and contraction $G \,/\, e$ (joining the two vertices).

For example:



(There is some sort of duality between deletion and contraction, as suggested by the symbols.)

Lemma 10 (Deletion-contraction for $t(G)$).

Given $e \in E$ (not a loop),

$$t(G) = t(G \setminus e) + t(G \,/\, e).$$

The first term is the number of spanning trees T where $e \notin T$, and the second term counts those where $e \in T$.

We can now continually apply deletion-contraction to compute $t(G)$. To speed this up, consider the following:

Definition (Isthmuses).

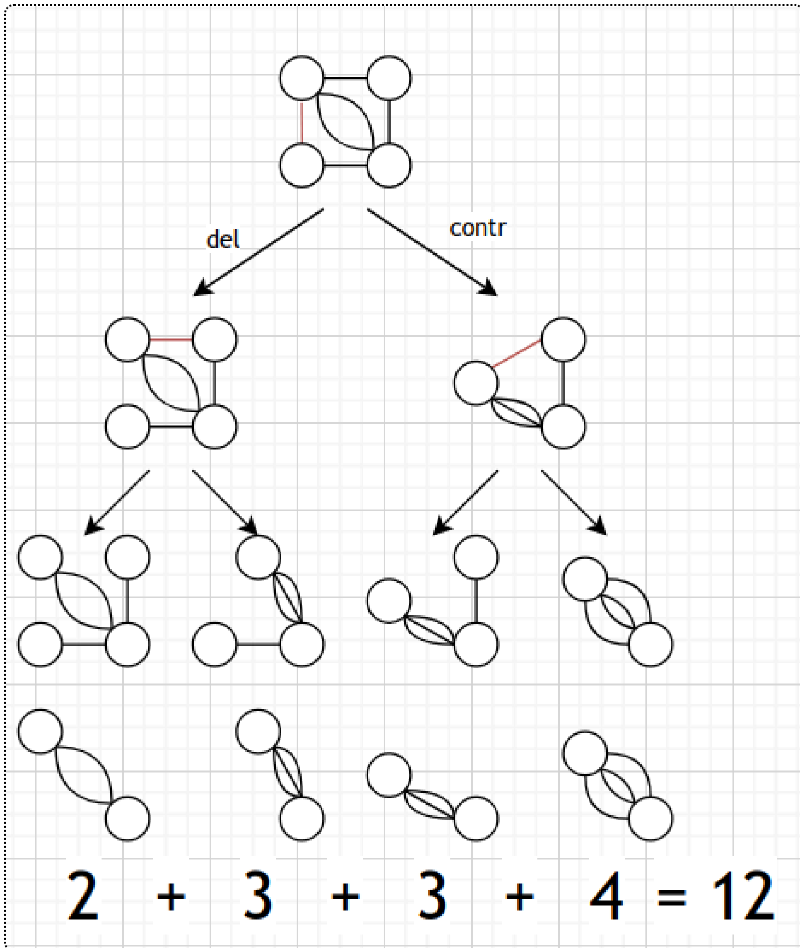
An edge $e \in E$ is called an **isthmus** (or a **bridge**) if $G \setminus e$ has more connected components than G.

Essentially, an isthmus connects two components. In particular, the edge from any leaf is an isthmus.

Lemma 11 ($t(G)$ with an isthmus).

If e is an isthmus, then $t(G) = t(G \,/\, e)$. (This is a special case of the above lemma.)

For instance, with our G above, we might do this:



From a computational point of view, deletion-contraction is a bit time-consuming (compared to listing all of the trees), but it's very helpful from a theoretical point of view, and it might help for some simple classes of graphs.

One method for counting spanning trees that works for arbitrary graphs is **Kirchhoff's matrix tree theorem**. We will go into it in more detail next class, but it involves several types of matrices relating to a tree:

- the *adjacency matrix*, a $|V| \times |V|$ square matrix, where (i, j) tells you how many edges there are between i and j.
- the *incidence matrix*, a $|V| \times |E|$ matrix, where (v, e) tells you if v belongs to e.

These two matrices are very different but are very easy to confuse! Be careful.

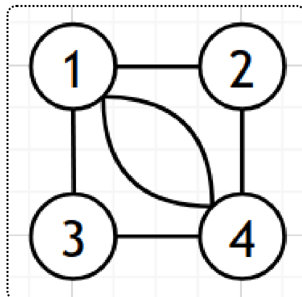
- the *Laplacian matrix*, a $|V| \times |V|$ matrix, a modification of the adjacency matrix where diagonal entries are changed.

Lecture 25 (April 4)

Kirchhoff's matrix tree theorem

Today we will discuss Kirchhoff's matrix tree theorem, named after the same Kirchhoff as Kirchhoff's circuit laws.

Let $G = (V, E)$ be a graph without loops (but possibly having multiple edges), like the following.



Then we have its adjacency matrix A (defined above); in this case

$$A = \begin{bmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 2 & 1 & 1 & 0 \end{bmatrix}.$$

The Laplacian L is a $n \times n$ matrix (where $n = |V|$)

$$L = \operatorname{diag}(d_1, \ldots, d_n) - A \qquad (d_i = \deg_G(i)).$$

For example, in our graph,

$$L = \begin{bmatrix} 4 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix} - A = \begin{bmatrix} 4 & -1 & -1 & -2 \\ -1 & 2 & 0 & -1 \\ -1 & 0 & 2 & -1 \\ -2 & -1 & -1 & 4 \end{bmatrix}.$$

Note that all sums of rows and columns are 0 (since the positive and negative terms cancel out), so the sum of all columns is a linear combination that adds to $\vec{0}$ and $\det L = 0$.

Now we are ready for the matrix tree theorem.

Theorem 12 (Kirchhoff's matrix tree theorem).

Fix $i \in [n]$. Let $\tilde{L}$ be a $(n-1) \times (n-1)$ matrix obtained by removing the ith row and column (called the reduced Laplacian, aka principal cofactors of L).

Then $t(G) = \det(\tilde{L})$ for any choice of i.

For example, select $i = 4$. Then

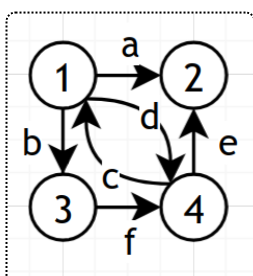
$$\tilde{L} = \begin{bmatrix} 4 & -1 & -1 \\ -1 & 2 & 0 \\ -1 & 0 & 2 \end{bmatrix}$$

and indeed $\det(\tilde{L}) = 12$.

Proof. We consider the *ordered incidence matrix* B, a $n \times m$ matrix (where $n = |V|$ and $m = |E|$):

- Arbitrarily direct all edges of G. Also, label every edge with a letter.
- In the column corresponding to each edge, place a 1 in the position corresponding to the origin vertex, -1 for the destination vertex, and 0 elsewhere.

For example:



$$B = \begin{bmatrix} 1 & 1 & -1 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 & -1 & 0 \\ 0 & -1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & -1 & 1 & -1 \end{bmatrix}$$

(where the columns are in a, b, c, d, e, f order.)

Lemma 13 (L and B).

$$L = B \cdot B^T.$$

Proof. $(B \cdot B^T)_{i,j}$ is the dot product of the ith and jth row of B.

If $i = j$, each edge that i is incident to contributes 1, and other edges contribute 0, so this is $\deg_G(i)$ as desired.
If $i \neq j$, each edge between i and j contributes -1, and other edges contribute 0, so this is $-(\# \text{ edges between } i \text{ and } j)$, as desired.

Similarly,

Lemma 14 ($\tilde{L}$).

$$\tilde{L} = \tilde{B} \cdot \tilde{B}^T,$$

where $\tilde{B}$ is the $(n-1) \times m$ matrix obtained by removing the ith row of B.

In order to use this to find $\det \tilde{L}$, we use the **Cauchy-Binet Formula**.

Theorem 15 (Cauchy-Binet Formula).

Suppose $k \leq m$, C is a $k \times m$ matrix, and D is a $m \times k$ matrix. Then

$$\underbrace{\det(C \cdot D)}_{k \times k \text{ matrix}} = \sum_{\substack{S \subset [m] \\ |S| = k}} \det(C_S) \det(D_S)$$

where C_S and D_S are $k \times k$ submatrices of C and D by taking the columns/rows in S.

For example, if

$$C = \begin{bmatrix} 1 & 2 & 3 \\ 6 & -6 & 5 \end{bmatrix}, \qquad D = \begin{bmatrix} \pi & e \\ 42 & \sqrt{2} \\ i & 0 \end{bmatrix},$$

then

$$\det(C \cdot D) = \begin{vmatrix} 1 & 2 \\ 6 & -6 \end{vmatrix} \cdot \begin{vmatrix} \pi & e \\ 42 & \sqrt{2} \end{vmatrix} + \begin{vmatrix} 1 & 3 \\ 6 & 5 \end{vmatrix} \cdot \begin{vmatrix} \pi & e \\ i & 0 \end{vmatrix} + \begin{vmatrix} 2 & 3 \\ -6 & 5 \end{vmatrix} \cdot \begin{vmatrix} 42 & \sqrt{2} \\ i & 0 \end{vmatrix}$$

The proof of this formula is left as an exercise, but there is a very nice one.

Now, we have $\det(\tilde{L}) = \det(\tilde{B} \cdot \tilde{B}^T)$, which is:

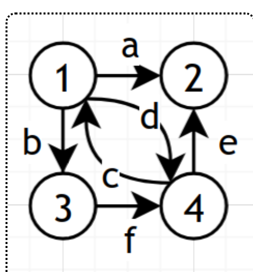
$$\sum_{\substack{S \subset [m] \\ |S| = n-1}} \det(\tilde{B}_S) \det(\tilde{B}_S^T) = \sum_{\substack{S \subset [m] \\ |S| = n-1}} (\det(\tilde{B}_S))^2$$

Essentially, the sum is selecting any $n - 1$ edges.

Lemma 16 $(\det \tilde{B}_S)$**.**

$$\det \tilde{B}_S = \begin{cases} \pm 1 & \text{if } S \text{ is the set of edges of a spanning tree of } G \\ 0 & \text{otherwise.} \end{cases}$$

This pretty clearly finishes the proof, so it suffices to prove this lemma. For example, in our original graph,



$S = \{a, b, c\}$ forms a spanning tree, and indeed

$$\begin{vmatrix} 1 & 1 & -1 \\ -1 & 0 & 0 \\ 0 & -1 & 0 \end{vmatrix} = 1.$$

But $\{a, d, e\}$ does not form a spanning tree, and

$$\begin{vmatrix} 1 & 1 & 0 \\ -1 & 0 & -1 \\ 0 & 0 & 0 \end{vmatrix} = 0.$$

Proof of Lemma. First, assume S does not form a spanning tree. Then the corresponding subgraph contains a cycle. Adding up the columns corresponding to the edges in the cycle (multiplying by -1 if necessary) yields the zero vector, so the columns of $\tilde{B}_S$ are not linearly independent and has determinant 0.

For example, suppose we had $a = (i_1 \to i_2)$, $b = (i_2 \to i_3)$, $c = (i_4 \to i_3)$, and $d = (i_4 \to i_1)$. Then the relevant portion of B looks like

$$\begin{bmatrix} 1 & & & -1 \\ -1 & 1 & & \\ & -1 & -1 & \\ & & 1 & 1 \end{bmatrix}$$

and indeed $\operatorname{col} a + \operatorname{col} b - \operatorname{col} c + \operatorname{col} d = \vec{0}$ (we subtract column c since its edge is oriented in the opposite direction.)

Now suppose S forms a spanning tree. Recall that we removed the ith row from B.

We can find a row with only one nonzero entry (since they correspond to leaves $\ell \neq i$ of the tree), so we can compute the determinant via minors using that row.

The determinant becomes $\pm$ the determinant for the subtree formed by removing ℓ and its edge from the tree. We repeat this process, removing leaves that are not i, until we get a 1×1 matrix corresponding to an edge (j, i) for some j. The determinant of this is ± 1, so the original determinant is also ± 1, as desired.

This concludes the proof of the matrix tree theorem.

Lecture 26 (April 7)

Here is another way to express the matrix tree theorem (recall L is the Laplacian matrix):

Corollary 17 (Eigenvalue version).

Let $\lambda_1, \ldots, \lambda_n$ be the eigenvalues of L, with $\lambda_n = 0$. Then

$$t(G) = \frac{1}{n}\lambda_1 \cdot \lambda_2 \cdot \ldots \lambda_{n-1}.$$

(This is a general fact about principal cofactors of matrices with rows/columns summing to 0.)

The core of the previous proof was the lemma about $\det \tilde{B}_S$. We will now present a closely related result.

Definition (Totally unimodular).

A matrix of B is **totally unimodular** if any minor of B (i.e. the determinant of a square submatrix of B) is either 0, 1, or -1.

Theorem 18 (Condition for total unimodularity).

Let B be any matrix such that every column of B is $\vec{e_i} - \vec{e_j}$, i.e. it only has two nonzero entries, one of which is 1 and the other is -1. Then B is totally unimodular.

As we proved before, a minor ends up being 0 if its submatrix contains a loop.

Examples of matrix tree theorem

Complete graph

Suppose $G = K_n$. Let's try to rederive Cayley's Formula. In this case,

$$L = \begin{bmatrix} n-1 & -1 & \cdots & -1 \\ -1 & n-1 & \cdots & \vdots \\ \vdots & \vdots & \ddots & -1 \\ -1 & \cdots & -1 & n-1 \end{bmatrix}.$$

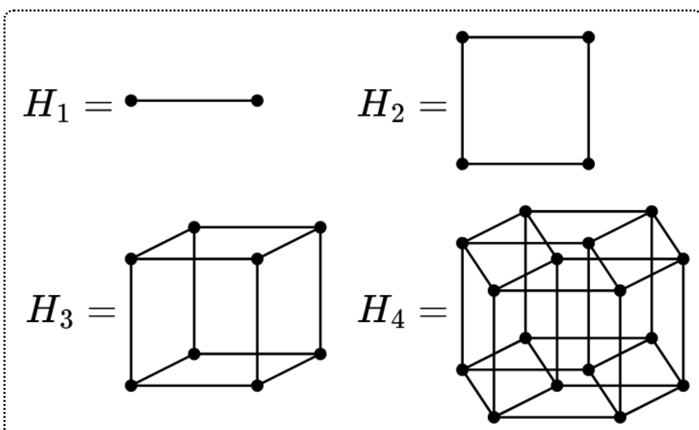
$\tilde{L}$ is the same, but as a $(n-1) \times (n-1)$ matrix. Let I_{n-1} be the $(n-1) \times (n-1)$ identity matrix, and J_{n-1} be the matrix containing all ones, so that $\tilde{L} = nI_{n-1} - J_{n-1}$.

If we can find eigenvalues of J_{n-1}, we can shift them to obtain eigenvalues of $\tilde{L}$ and therefore $t(G)$. The rank of J_{n-1} is 1, so there is only one nonzero eigenvalue, which is $n-1$ (corresponding to $(1, 1, 1, \ldots, 1)$).

As such, the eigenvalues of $\tilde{L}$ are $n - (n-1) = 1$ and $n-2$ copies of $n - 0 = n$. So, $t(G) = \det(\tilde{L})$ is the product of these eigenvalues, i.e. n^{n-2} as desired.

Hypercube graph

Let H_d be the skeleton of a d-dimensional cube. For example:



By observation, $t(H_1) = 1$, $t(H_2) = 4$. What is $t(H_3)$?

The answer is 384. This sequence grows very quickly! We want to find an explicit formula.

First, note that H_d is a d-regular graph, i.e. every vertex has degree d. This makes it slightly easier to apply the matrix tree theorem, since then $L = dI - A$

and we only have to find the eigenvalues of the adjacency matrix: if A has eigenvalues $\alpha_1, \ldots, \alpha_n$, L has eigenvalues $d - \alpha_1, d - \alpha_2, \ldots, d - \alpha_n$ (like we did with K_n).

To construct hypercube graphs, we will need the operation of creating products of graphs:

Given $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$, $G_1 \times G_2 = (V, E)$, where

$$V = V_1 \times V_2 = \{(u, v) \mid u \in V_1, v \in V_2\}$$

and

$$E = \{(u, v), (u', v') \mid \left[(u, u') \in E_1 \text{ and } v = v_1\right] \text{ or } \left[(u, u') \in E_1 \text{ and } v = v_1\right]\}$$

We can see that $H_d = H_{d-1} \times H_1$.

Lemma 19 ($A(G \times H)$).

Suppose $A(G)$ has eigenvalues $\alpha_1, \alpha_2, \ldots, \alpha_m$, and $A(H)$ has eigenvalues $\beta_1, \beta_2, \ldots, \beta_n$.

Then the eigenvalues of $A(G \times H)$ are $\alpha_i + \beta_j$ for all $i \in [m]$, $j \in [n]$.

Note that indeed we have $m \cdot n$ eigenvalues, as expected. The proof of this lemma is left as a linear algebra exercise by combining eigenvectors, but here's an example:

Suppose G comprises the edge $(1, 2)$ and H comprises the edge (a, b). Then $G \times H$ is a square with vertices $1a$, $1b$, $2a$, $2b$. $A(G) = A(H) = \left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right]$, which has eigenvalues 1 (for $\left[\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right]$) and -1 (for $\left[\begin{smallmatrix} 1 \\ -1 \end{smallmatrix}\right]$). Meanwhile,

$$A(G \times H) = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

The eigenvectors are:

$$\begin{bmatrix}1\\1\\1\\1\end{bmatrix}, \begin{bmatrix}1\\1\\-1\\-1\end{bmatrix}, \begin{bmatrix}1\\-1\\1\\-1\end{bmatrix}, \begin{bmatrix}1\\-1\\-1\\1\end{bmatrix}$$

They have corresponding eigenvalues $1+1$, $1+(-1)$, $(-1)+1$, $(-1)+(-1)$, as desired.

Due to this lemma, $A(H_d)$ has eigenvalues $\pm 1 \pm 1 \pm 1 \cdots \pm 1$ (where we have 2^d choices of signs.) We can see that it will have eigenvalue $d - 2k$ with multiplicity $\binom{d}{k}$ (by picking the k ones to turn negative). As such, $L(H_d) = dI - A$ has eigenvalue $2k$ with multiplicity $\binom{d}{k}$, and:

> **Theorem 20** ($t(H_d)$).
>
> $$t(H_d) = \frac{1}{2^d} \prod_{k=1}^{d} (2k)^{\binom{d}{k}} = 2^{2^d - d - 1} \prod_{k=1}^{d} k^{\binom{d}{k}}.$$

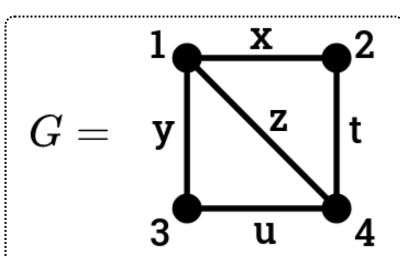
Lecture 27 (April 9)

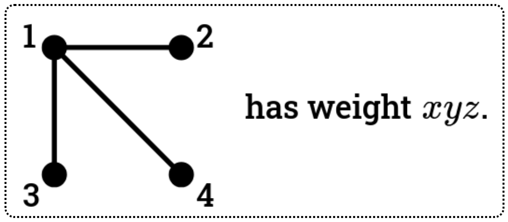
Today, we will discuss two orthogonal generalizations of the matrix tree theorem (MTT), namely weighted and directed versions.

Weighted MTT

As usual, we take a graph $G = (V, E)$ on $V = [n]$. We assign a weight $\mathrm{wt}(e)$ to each edge. We then want to find the weighted sum of spanning trees T, where we define $\mathrm{wt}(T) := \prod_{e \in T} \mathrm{wt}(e)$.

For example:





We can also define an adjacency matrix $A = (a_{ij})$ for a weighted graph. We simply set a_{ij} to the weight of the edge between i and j. (An edge of weight 0 is equivalent to not having an edge in this use case.) For instance, the above graph has the adjacency matrix

$$A = \begin{bmatrix} 0 & x & y & z \\ x & 0 & 0 & t \\ y & 0 & 0 & u \\ z & t & u & 0 \end{bmatrix}$$

Note that we can take any arbitrary symmetric matrix with 0s along the diagonal as an adjacency matrix.

We can define the Laplacian similarly, where $L = \operatorname{diag}(d_1, \ldots, d_n) - A$ and $d_i = \sum_{j \neq i} a_{ij}$. So L is an arbitrary symmetric $n \times n$ matrix such that columns add to 0.

Now, we can present the weighted matrix tree theorem:

Theorem 21 (Weighted MTT)**.**

Let $\tilde{L}$ be any principal cofactor of L, a $(n-1) \times (n-1)$ matrix. Then

$$\sum_{T \text{ spanning tree}} \operatorname{wt}(T) = \det(\tilde{L}).$$

Recall that about a week ago we defined a different notion of the weight of a graph, a polynomial

$$\operatorname{wt}^{\text{old}}(T) := \prod_{i=1}^{n} x_i^{\deg_T(i)-1}.$$

We can relate these two definitions: if $\mathrm{wt}(i \leftrightarrow j) = x_i \cdot x_j$, then

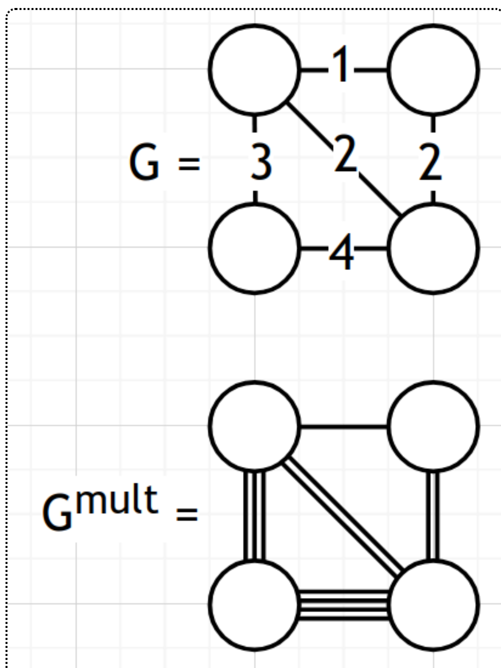
$$\mathrm{wt}(\mathrm{T}) = (x_1 \dots x_n)\mathrm{wt}^{\mathrm{old}}(T).$$

So, as an exercise, we can deduce the multivariate Cayley formula from the weighted MTT.

Proof. Weighted MTT follows pretty easily from the original matrix-tree theorem. We can again use the trick where proving the equality on all positive integers implies that the polynomials are identically equal.

If all edge weights are positive integers, we can transform our graph G into a new unweighted graph by using multiple edges: replace an edge $(i \leftrightarrow j)$ with weight w with w edges $(i \leftrightarrow j)$.

For example:

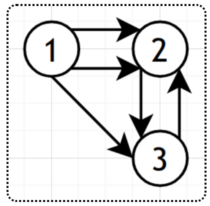


Now, we can see that $\sum_{T \subset G} \mathrm{wt}(\mathrm{T})$ is just the number of spanning trees of G^{mult}: the weight of a tree is just the number of ways to pick which of the parallel edges to use in a spanning tree of G^{mult}.

Therefore, the weighted matrix tree theorem is just equivalent to our usual MTT on the graph G^{mult}, so it is true for positive integer weights, and therefore the polynomials $\sum \mathrm{wt}(\mathrm{T})$ and $\det(\tilde{L})$ are equal, as desired.

Directed MTT

Let $G = (V, E)$ be a directed graph, like the following:



We need a new notion of "tree". To do this, we fix a root vertex $r \in V$.

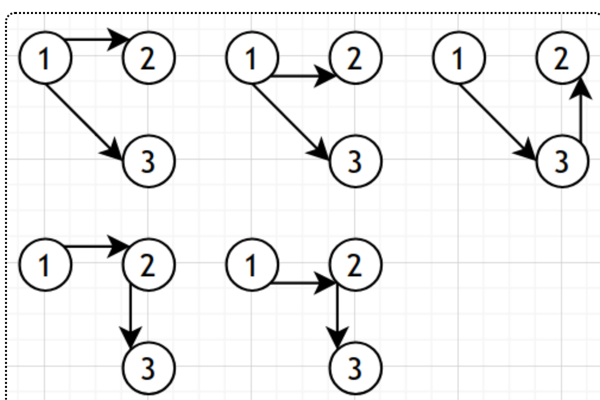
Definition (Arborescence).

An **arborescence** T of G is a subgraph of G such that:

1. T is a spanning tree of G as an undirected graph.
2. For all $v \in V$, there exists a directed path in T from r to v.

These end up looking like trees, but with each edge being directed away from r. We define $\mathrm{Arb}_r(G)$ as the number of arborescences with root r.

In our example, $\mathrm{Arb}_1(G) = 5$ (seen below), $\mathrm{Arb}_2(G) = 0$, and $\mathrm{Arb}_3(G) = 0$.



Once again, we need to redefine the adjacency and Laplacian matrices. The adjacency matrix $A = (a_{ij})$ will have the number of directed edges $i \to j$ in a_{ij}. The Laplacian matrix $L = \mathrm{diag}(d_1, \ldots, d_n) - A$ will have d_i being the indegree of i in G.

In our example:

$$A = \begin{bmatrix} 0 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \qquad L = \begin{bmatrix} 0 & -2 & -1 \\ 0 & 3 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

L still has columns adding to 0, but rows may not necessarily add to 0.

Recall that the cofactors L^{ij} of L are defined as:

$$L^{ij} = (-1)^{i+j} \cdot \det(L \text{ without } i\text{th row and } j\text{th column}).$$

Then,

Theorem 22 (Directed MTT).

Fix any $r, k \in [n]$. Then

$$\mathrm{Arb}_r(G) = L^{kr}.$$

For example, with $r = 1$ in our above graph, we obtain the following cofactors:

$$L^{11} = \begin{vmatrix} 3 & -1 \\ -1 & 2 \end{vmatrix} = 5 \qquad L^{21} = -\begin{vmatrix} -2 & -1 \\ -1 & 2 \end{vmatrix} = 5 \qquad L^{31} = \begin{vmatrix} -2 & -1 \\ 3 & -1 \end{vmatrix} = 5.$$

Note that the usual MTT is a special case of the directed MTT, for directed graphs G such that $a_{ij} = a_{ji}$: once we fix a root r and a spanning tree, the directions of the corresponding arborescence are forced (as all edges point away from r). This actually means that for an undirected graph, we can use any cofactor rather than only principal ones.

There are two very nice proofs of the directed MTT:

- using induction on $|E|$
- using the involution principle (which we previously used for the pentagonal number theorem).
  Both are actually much simpler (easier and shorter) than the Cauchy-Binet proof of MTT: for the inductive proof, for example, allowing directed graphs gives us a stronger induction hypothesis that makes it possible. We will fully go over them in the next few classes.

Lecture 28 (April 11)

(I am out of class; thanks again to Annie Guo)

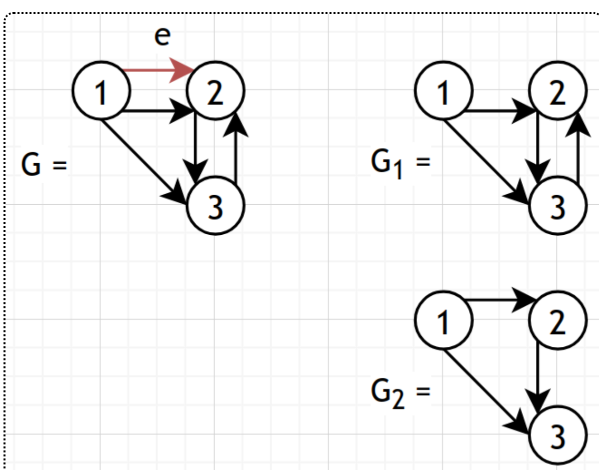
Today we will fully prove the directed matrix-tree theorem, $\mathrm{Arb}_r(G) = L^{kr}$. We will induct on the number of edges, $|E|$.

Base case: All edges in G end at the root r. Then if $n \geq 2$, $\mathrm{Arb}_r(G) = 0 = L^{kr}$, and if $n = 1$ both are equal to 1 (determinant of a 0×0 matrix is 1 by convention.)

Inductive step: The crucial lemma is the following:

> **Lemma 23** (Adapted deletion-contraction).
>
> Let $e = (i \to j)$ be an edge of G such that $j \neq r$. Define
>
> $$G_1 = G \setminus e$$
> $$G_2 = G \setminus \{f \mid f = (k, j) \neq e\} \text{ (i.e. } f \text{ and } e \text{ have the same endpoint)}$$
>
> Then
>
> $$\mathrm{Arb}_r(G) = \mathrm{Arb}_r(G_1) + \mathrm{Arb}_r(G_2).$$
>
> If $G_2 = G$, i.e. e is the only edge ending at j, then $\mathrm{Arb}_r(G) = \mathrm{Arb}_r(G \,/\, e)$.

Here's an example.



This lemma is true because for any arborescence and vertex j, there will be exactly one edge f entering j. If $f = e$, it's part of $\mathrm{Arb}_r(G_2)$, and otherwise it's in

$\mathrm{Arb}_r(G_2)$. In the case where $G_2 = G$, we are forced to include e, so we can just contract that edge.

Now let's take a look at their Laplacians. They are very similar, only differing in the jth column, and they have an additive relationship (so their cofactors will also be additive):

$$L_G = \begin{bmatrix} 0 & \mathbf{-2} & -1 \\ 0 & \mathbf{3} & -1 \\ 0 & \mathbf{-1} & 2 \end{bmatrix}$$
$$L_{G_1} = \begin{bmatrix} 0 & \mathbf{-1} & -1 \\ 0 & \mathbf{2} & -1 \\ 0 & \mathbf{-1} & 2 \end{bmatrix}$$
$$L_{G_2} = \begin{bmatrix} 0 & \mathbf{-1} & -1 \\ 0 & \mathbf{1} & -1 \\ 0 & \mathbf{0} & 2 \end{bmatrix}$$

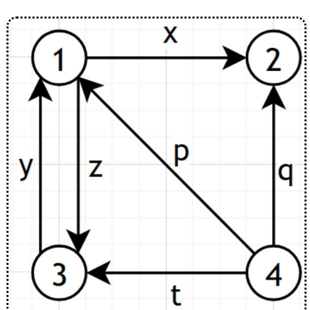
This is evident by considering what every non-diagonal entry of the jth column represents: the count of edges that end at j. One can also that contracting e in the case where $G = G_2$ does not affect the cofactors of the Laplacian.

Now, since we can remove all edges not ending at r, we can reduce any graph to the base case and finish.

The second proof (via involution) will be fully explained next class. This proof works for the special case where we take principal cofactors, i.e. $r = k$. The idea is to expand the determinant into individual terms (for each permutation of the rows) which we analyze using cycle notation.

Lecture 29 (April 14)

We will go through the proof using the involution principle today. We will use this example:



Recall that we are trying to prove

$$\sum_{\substack{\text{arborescence}\\ \text{with root } n}} \mathrm{wt}(T) = L^{nn},$$

assuming that the root is the maximal vertex. We expand the cofactor (by considering all permutations of $[n-1]$):

$$L^{nn} = \sum_{w \in S_{n-1}} (-1)^{\mathrm{inv}(w)} \prod_{i=1}^{n-1} \ell_{i,w(i)}.$$

For any given permutation w, note that each cycle or fixed point (in the permutation sense) is self-contained. If a cycle of the permutation is not a cycle in the graph, the product will just resolve to 0, so we can ignore those. So this becomes

$$\sum (-1)^{\#\text{ cycles}} \left(\prod_{\substack{e \text{ edge}\\ \text{in cycle}}} {\color{green} x_e} \right) \left(\prod_{\substack{i \text{ fixed}\\ \text{point}}} \sum_{\substack{f \text{ edge}\\ \to i}} {\color{blue} x_f} \right).$$

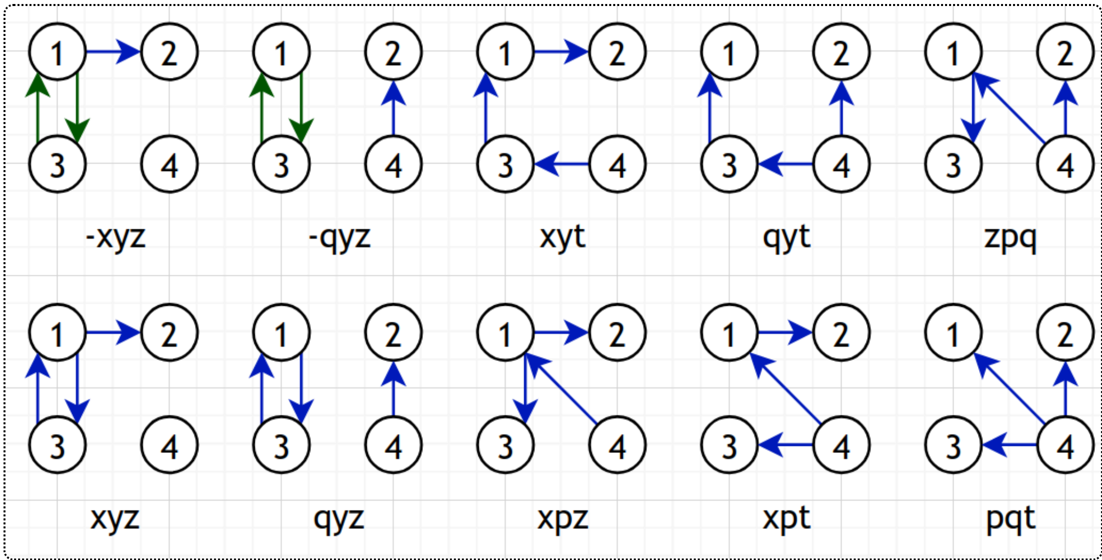
The sign comes from combining the sign of the permutation $(-1)^{\mathrm{inv}(w)}$ with the the fact that weights in off-diagonal entries of the Laplacian are negated. We see that any cycle of length k contributes a factor of -1 (since the sign of the cycle itself is $(-1)^{k-1}$, and we multiply by $(-1)^k$ from the k edges), hence $(-1)^{\#\text{ cycles}}$.

After expanding the inner sum, what each term essentially represents is a coloring of some edges $H \subset G$ to be either green or blue such that:

- all green edges form a disjoint union of directed cycles (length at least 2)
- for all vertices $i \in [n-1]$ not in a green cycle, exactly one blue edge enters vertex i (select one from the inner sum).

Then, we sum the product of weights in H, $\mathrm{wt}(H) = \prod_{e \in H} x_e$, multiplied by the sign $(-1)^{\#\text{ green cycles}}$.

In our example, we have ten options:



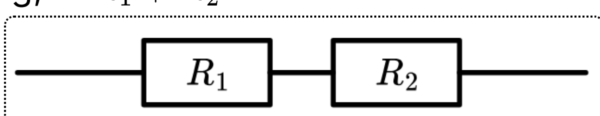
Note how all colorings with green cycles and those with blue cycles neatly cancel out (we can take an involution that preserves weight but reverses sign: flip the color of the lexicographically first cycle.)

That leaves only the graphs with only blue edges that have no directed cycles, which are precisely arborescences rooted at n (since no edge enters n), and their weights are added up as desired.

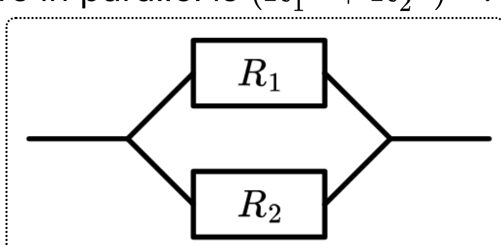
Application (Electrical network)

Suppose we have an electrical network with a bunch of resistors.

You may know from physics class that the resistance of some resistors in series, like the following, is $R_1 + R_2$:

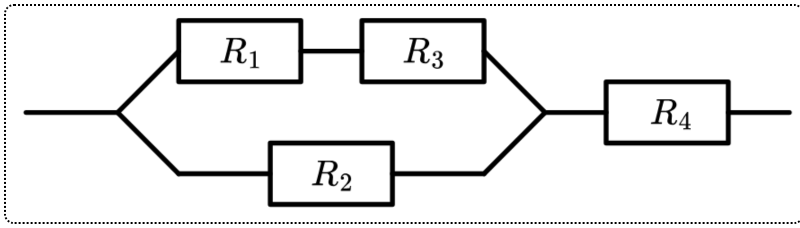


The resistance of resistors in parallel is $(R_1^{-1} + R_2^{-1})^{-1}$:

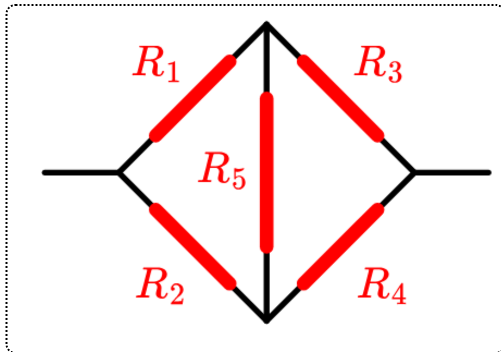


We can combine these rules to find the resistance of more complicated circuits like the following, which has effective resistance

$$\frac{1}{\frac{1}{R_1+R_2}+\frac{1}{R_3}}+R_4:$$



But doing this is impossible for some circuits, like the **Wheatstone bridge**:



In the special case where $R_1 = R_2 = R_3 = R_4 = R_5$, by considering symmetry, we realize that the electric potential of the two endpoints of R_5 are equal so we can merge those into one point.

Let's try to formalize this into a math problem. We represent our network as a graph G with two selected vertices A and B, where each edge e with resistance R_e has weight $\frac{1}{R_e}$. Then, we claim that

Theorem 24 (Effective resistance)**.**

The effective resistance between A and B, $R_{AB}(G)$, is

$$\frac{\sum\limits_{\substack{T' \text{ spanning} \\ \text{tree of } G'}} \operatorname{wt}(T')}{\sum\limits_{\substack{T \text{ spanning} \\ \text{tree of } G}} \operatorname{wt}(T)}$$

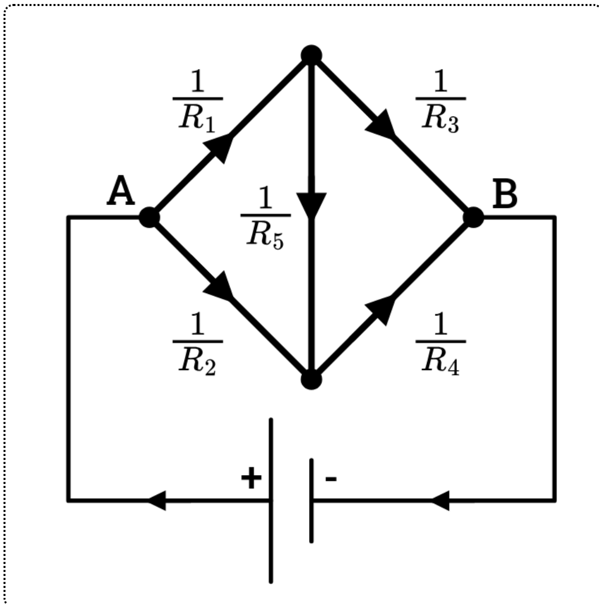
where G' is obtained from G by gluing A and B.

In the next lecture, we will prove how this theorem follows from the rules of electricity.

Lecture 30 (April 16)

Our graph representation is undirected, but in order to talk about currents, we need to pick some edge directions, but they can be completely arbitrary.

Here's what our graph might look like (recall we want to find the resistance from A to B, so we attach a battery between them):



For every edge $e \in E$, we have a resistance $R_e > 0$, a current $I_e \in \mathbb{R}$, and a voltage $V_e \in \mathbb{R}$. If we reverse the direction of e to obtain $-e$, note that $R_{-e} = R_e$, $I_{-e} = -I_e$, and $V_{-e} = -V_e$.

We need three laws of electricity:

- *1st Kirchhoff Law*: For all vertices, the incoming current equals the outgoing current.
- *2nd Kirchhoff Law*: For all cycles C in G, the signed sum of voltages of edges in C is 0 (i.e. take a voltage to be positive if the edge direction agrees with the cycle, take it to be negative if it is oppositely directed.).

Note that there is a duality between these two laws (using the dual operation between planar graphs.)

- *Ohm's Law*: For any edge $e \in E$, $V_e = I_e \cdot R_e$.

In our problem, we know the voltage of the battery as well as the resistance of each edge, and we want to use the linear equations resulting from these laws to determine the total effective resistance.

Lemma 25 (Potential function).

The 2nd Kirchhoff law is equivalent to there being a potential function $U : V \to \mathbb{R}$ such that for all edges $e = (u \to v)$, $V_e = U_v - U_u$.

So, let's create a vector of potentials.

$$\vec{U} = \begin{bmatrix} U_1 \\ U_2 \\ \vdots \\ U_n \end{bmatrix}$$

Then, we rewrite Ohm's law as defining

$$I_e = \frac{U_v - U_u}{R_e}$$

and now we only need to represent the Kirchhoff's 1st law in a matrix form.

Suppose a vertex v has some incoming edges from v_1, v_2, v_3 and outgoing edges to v_4, v_5, with currents $I_1 \dots I_5$. We know that $I_1 + I_2 + I_3 = I_4 + I_5$, and by expanding each I_e we get

$$\frac{U_v - U_{v_1}}{R_1} + \frac{U_v - U_{v_2}}{R_2} + \frac{U_v - U_{v_3}}{R_3} = \frac{U_{v_4} - U_v}{R_4} + \frac{U_{v_5} - U_v}{R_5}.$$

Note that if we move all terms to one side, the expression doesn't depend on edge directions at all, so

$$\sum_{i=1}^{5} \frac{U_v - U_{v_i}}{R_i} = 0.$$

Let's combine all of the U_v terms together. For a general vertex v of degree d, connected to v_i with resistance R_i, we see

$$\left(\frac{1}{R_1} + \dots + \frac{1}{R_d}\right) U_v - \sum_{i=1}^{d} \frac{1}{R_i} \cdot U_{v_i} = \begin{cases} 0 & i \neq A, B \\ I_{bat} & i = A \qquad \text{(connected to + terminal)} \\ -I_{bat} & i = B \qquad \text{(connected to − terminal)} \end{cases}$$

This equation can be rewritten in terms of the "Kirchhoff matrix" K as applied to $\vec{U}$, which is really the exact same as the Laplacian for a graph with edge weight $\frac{1}{R_e}$: each entry K_{ij} will be

$$\begin{cases} -\frac{1}{R_e} & \text{if } i \neq j \text{ are connected by an edge} \\ \sum \frac{1}{R_e} \text{ (over all edges incident to } i) & \text{if } i = j \end{cases}.$$

So, we need to solve

$$K \cdot \vec{U} = \begin{bmatrix} I_{bat} \\ 0 \\ \vdots \\ 0 \\ -I_{bat} \end{bmatrix}$$

(assume that our vertex A is 1 and B is n). We need to fix it somehow (since we can shift all potentials by any value) so we "ground" the negative terminal of the battery, i.e. setting $U_n = 0$. Then we end up with a system of $n-1$ equations,

$$\tilde{K} \cdot \begin{bmatrix} U_1 \\ \vdots \\ U_{n-1} \end{bmatrix} = \begin{bmatrix} I_{bat} \\ 0 \\ \vdots \\ 0 \end{bmatrix},$$

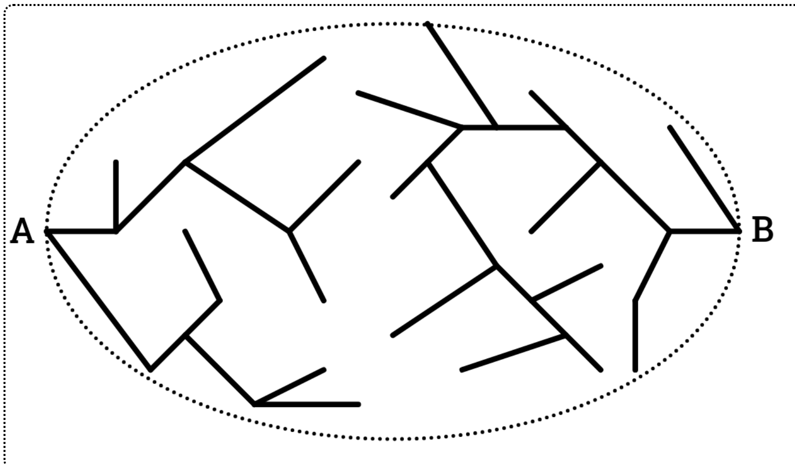
where $\tilde{K}$ is exactly the reduced Laplacian (obtained by removing the nth row and column). We know that the voltage of the battery will be $U_1 - U_n = 0$, so the effective resistance is just U_1 / I_{bat}. We can rescale and assume that I_{bat}.

By Cramer's rule from linear algebra, $U_1 = \frac{\det(\tilde{K}_1)}{\det(\tilde{K})}$, where $\tilde{K}_1$ is obtained by replacing the first column of $\tilde{K}$ with $\begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$. Its determinant is equal to that of the $(n-2) \times (n-2)$ matrix obtained by removing the first and last row and column of K.

What does this mean in terms of our graph? It ends up corresponding to the action of gluing the first and last vertices together, so the effective resistance is

$$R_{1n}(G) = \frac{\sum\limits_{F \text{ a 2-forest}} \prod\limits_{e \in F} \frac{1}{R_e}}{\sum\limits_{T \text{ a spanning tree}} \prod\limits_{e \in T} \frac{1}{R_e}},$$

where a 2-forest is a forest with exactly two connected components such that our marked vertices ($A = 1, B = n$) are in different connected components. (Essentially, we grow one tree from each terminal of the battery, and cover all vertices of G).

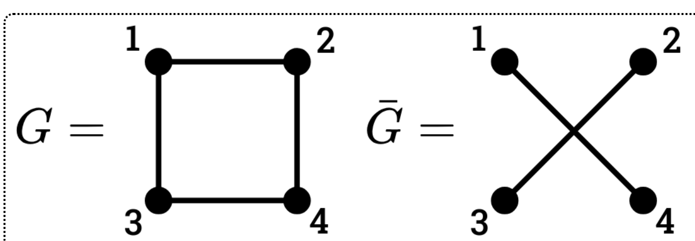


For example, in our Wheatstone bridge, we have eight 2-forests with two edges and eight spanning trees, so it looks like

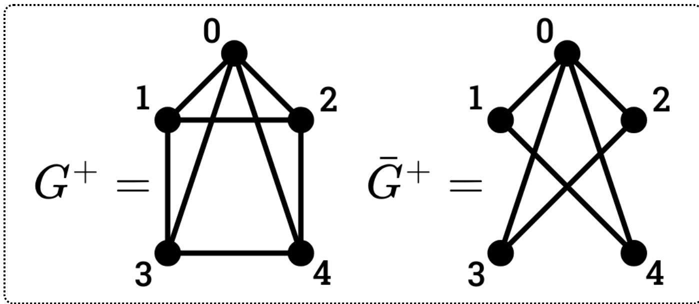
$$\frac{\frac{1}{R_1}\frac{1}{R_2} + \frac{1}{R_1}\frac{1}{R_5} + \dots}{\frac{1}{R_1}\frac{1}{R_2}\frac{1}{R_3} + \frac{1}{R_1}\frac{1}{R_2}\frac{1}{R_4} + \dots}$$

Lecture 31 (April 18)

Let $G = (V, E)$ be a simple graph on $V = [n]$ (i.e. it has no multiple edges or loops). We define the *complementary graph* $\overline{G} = (V, \overline{E})$ as the graph containing an edge $(i, j) \in \overline{E}$ iff $(i, j) \notin E$.



Is there a relationship between spanning trees of G and those of $\overline{G}$? This would be slightly surprising, and the easy answer is no. But what if we add a root vertex 0, so that $G^+ = (V^+, E^+)$, where $V^+ = \{0, \ldots, n\}$ and $E^+ = E \cup \{(0,i) \mid i \in [n]\}$?



Then, every spanning tree T of G^+ corresponds to a rooted forest F of G (where in each connected component of F, we designate one vertex as the root), by removing the edges from 0 in the tree.

We consider the generating function $F_G(x)$ that represents the number of rooted forests with some number of connected components. This is then

$$F_G(x) = \sum_{\substack{T \text{ spanning} \\ \text{tree of } G^+}} x^{\deg_T(0)-1}.$$

The -1 is there since there clearly cannot be no connected components. In our G above, we can verify that

$$F_G(x) = x^3 + (4 \cdot 2)x^2 + (2 \cdot 2 \cdot 2 + 4 \cdot 3)x + 4 \cdot 4 = x^3 + 8x^2 + 20x + 16.$$

Note that the constant term is n times the number of spanning trees of G, for example, since each spanning tree can be rooted at any of n vertices, and the leading term is always x^{n-1} (since it represents a forest with all isolated vertices).

For the complementary graph, it turns out that $F_{\overline{G}}(x) = x^3 + 4x^2 + 4x$. There is in fact a relation:

Theorem 26 (Reciprocity theorem (Bedrosian 1964)).

$$F_G(x) = (-1)^{n-1} F_{\overline{G}}(-n-x)$$

For example, $x^3 + 8x^2 + 20x + 16 = (-1)^3((-x-4)^3 + 4(-x-4)^2 + (4(-x-4)))$.

The proof of this theorem will be left as an exercise (if you want to use it on a problem set problem, you will have to provide a proof). Some possible strategies include directly deducing it from the matrix tree theorem and bijecting using the following multivariate generalization: for

$$F_G(x_0; x_1, x_2, \cdots, x_n) := \sum_{\substack{T \text{ spanning} \\ \text{tree of } G^+}} \prod_{i=0}^{n} x_i^{\deg_T(i)-1},$$

we have

$$F_G(x_0; x_1, \cdots, x_n) = (-1)^{n-1} F_{\overline{G}}(-x_0 - x_1 - \cdots - x_n; x_1, \cdots, x_n).$$

For $G = K_3$, both sides equal $(x_0 + x_1 + x_2 + x_3)^3$ (same as multivariate Cayley's Formula on a K_4).

Suppose G_1 and G_2 are on two disjoint sets of vertices V_1 and V_2. Consider their disjoint union $G_1 \sqcup G_2$. Then, the spanning trees T of $(G_1 \sqcup G_2)^+$ can be bijected with pairs (T_1, T_2) of spanning trees of G_1^+ and G_2^+. Then

$$\deg_T(0) - 1 = (\deg_{T_1}(0) - 1) + (\deg_{T_2}(0) - 1) + 1,$$

so $F_{G_1 \sqcup G_2}(x) = x F_{G_1}(x) \cdot F_{G_2(x)}$.

We can now use these three identities to compute spanning trees in some graphs. $F_{K_1}(x) = 1$, so we can get Cayley's Formula: letting O_n denote the graph on n vertices with no edges (i.e. $\overline{K}_n$),

$$\begin{aligned} F_{O_n}(x) &= x(x(\ldots(F_{K_1}(x))\ldots)F_{K_1}(x))F_{K_1}(x) \\ &= x^{n-1} \end{aligned}$$

$$\begin{aligned} t(K_n) &= \frac{1}{n} F_{K_n}(0) \\ &= \frac{1}{n}(-1)^{n-1} F_{O_n}(-n-0) \\ &= \frac{1}{n}(-1)^{n-1}(-n)^{n-1} \\ &= \frac{1}{n}(n)^{n-1} = n^{n-2}. \end{aligned}$$

Similarly, for the complete bipartite graph $K_{m,n} = \overline{K_m \sqcup K_n}$, we now have

$$t(K_{m,n}) = \frac{1}{m+n}(n+m) \cdot n^{m-1}m^{n-1}.$$